

Volume Vi Reference

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Introduction to Algebraic Structures

0.1 What Is a Mathematical Structure?

0.1.1 Algebraic Structures

Definition 0.1.1 (Algebraic Structure). An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A : each n -ary function symbol as an operation $A^n \rightarrow A$, each n -ary relation symbol as a subset of A^n , and each constant symbol as an element of A . The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Definition 0.1.2 (First-Order Signature). A **first-order signature**

$$\mathcal{L} = (\text{Const}_{\mathcal{L}}, \text{Func}_{\mathcal{L}}, \text{Rel}_{\mathcal{L}}, \text{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Definition 0.1.3 ($\mathcal{L}$ -Structure). Given a signature $\mathcal{L}$, an **$\mathcal{L}$ -structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A .

0.1.2 Laws of Composition

Definition 0.1.4 (Binary Operation / Law of Composition). Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \rightarrow A.$$

The image of (a, b) is written $a \star b$.

Definition 0.1.5 (n -ary Operation). An **n -ary operation** on A is a function $f : A^n \rightarrow A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Definition 0.1.6 (Unary Operation). A **unary operation** on A is a function $f : A \rightarrow A$.

Definition 0.1.7 (Nullary Operation / Constant). A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \rightarrow A$.

0.1.3 Distinguished Elements

Definition 0.1.8 (Left Identity). An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 0.1.9 (Right Identity). An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 0.1.10 (Identity / Neutral Element). An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Proposition 0.1.11 (Identity Collapse and Uniqueness). *If $\star$ has a left identity e_L and a right identity e_R , then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique. Go to proof.*

Definition 0.1.12 (Left Absorbing Element). An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 0.1.13 (Right Absorbing Element). An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 0.1.14 (Absorbing Element). An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Proposition 0.1.15 (Uniqueness of the Absorbing Element). *A two-sided absorbing element, if it exists, is unique. Go to proof.*

0.1.4 Inverses

Definition 0.1.16 (Left Inverse). Let $\star$ have identity e , and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 0.1.17 (Right Inverse). Let $\star$ have identity e , and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 0.1.18 (Inverse). Let $\star$ have identity e , and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Definition 0.1.19 (Invertible Element). An element a is **invertible** (a unit) if it has a two-sided inverse.

Proposition 0.1.20 (Inverse Collapse and Uniqueness in a Monoid). *If $\star$ is associative with identity e , and a has a left inverse b and a right inverse c , then $b = c$. Hence the inverse of an invertible element is unique. Go to proof.*

Definition 0.1.21 (Additive Inverse). When $\star = +$ with identity 0 , the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 0.1.22 (Multiplicative Inverse). When $\star = \cdot$ with identity 1 , the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 0.1.23 (Units of a Monoid Form a Group). *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group. Go to proof.*

0.1.5 Properties of Laws

Definition 0.1.24 (Associativity). A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Theorem 0.1.25 (General Associativity). *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization. Go to proof.*

Definition 0.1.26 (Powers / Iterated Operation). For associative $\star$ with identity e , define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 0.1.27 (Commutativity). A binary operation $\star$ on A is **commutative** if $a \star b = b \star a$ for all $a, b \in A$.

Definition 0.1.28 (Left Distributivity). For laws $+$ and $\cdot$ on A , multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 0.1.29 (Right Distributivity). For laws $+$ and $\cdot$ on A , multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 0.1.30 (Distributivity). For laws $+$ and $\cdot$ on A , multiplication is **distributive** over addition if it is both left distributive and right distributive.

Definition 0.1.31 (Idempotent Operation). A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

0.1.6 Regularity Properties

Definition 0.1.32 (Left Cancellable Element). An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \implies x = y$$

for all $x, y \in A$.

Definition 0.1.33 (Right Cancellable Element). An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \implies x = y$$

for all $x, y \in A$.

Definition 0.1.34 (Cancellable Element). An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 0.1.35 (Invertible Implies Cancellable). *In a monoid, every invertible element is cancellable. Go to proof.*

Definition 0.1.36 (Zero Divisor). In a ring R , a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

0.1.7 One Internal Law

Definition 0.1.37 (Magma). A **magma** is a pair $(A, \star)$ where $A \neq \emptyset$ and $\star$ is a binary operation on A .

Definition 0.1.38 (Semigroup). A **semigroup** is a magma whose operation is associative.

Definition 0.1.39 (Monoid). A **monoid** is a semigroup with an identity element.

Definition 0.1.40 (Group). A **group** is a monoid in which every element has an inverse.

Definition 0.1.41 (Abelian Group). An **abelian group** is a group whose operation is commutative.

Definition 0.1.42 (Order of a Group). The **order** of a group G , denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 0.1.43 (Uniqueness of the Identity in a Group). *The identity of a group is unique. Go to proof.*

Proposition 0.1.44 (Uniqueness of Inverses in a Group). *Each element of a group has a unique inverse. Go to proof.*

Proposition 0.1.45 (Cancellation Laws in a Group). *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$. Go to proof.*

Proposition 0.1.46 (Socks-Shoes). *In a group, $(ab)^{-1} = b^{-1}a^{-1}$. Go to proof.*

0.1.8 Two Internal Laws

Definition 0.1.47 (Rng). A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 0.1.48 (Ring). A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 0.1.49 (Commutative Ring). A **commutative ring** is a ring whose multiplication is commutative.

Definition 0.1.50 (Integral Domain). An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 0.1.51 (Division Ring / Skew Field). A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 0.1.52 (Field). A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 0.1.53 (Multiplication by Zero). *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a . Go to proof.*

Proposition 0.1.54 (Multiplication by Additive Inverse). *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$. Go to proof.

Proposition 0.1.55 (Cancellation in Integral Domains). *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$. Go to proof.*

Proposition 0.1.56 (Every Field is an Integral Domain). *Every field is an integral domain. Go to proof.*

Proposition 0.1.57 (Finite Integral Domain is a Field). *Every finite integral domain is a field. Go to proof.*

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Abstract Algebra

0.13 Induction

0.13.1 Induction

Definitions and Theorems

0.14 Integers

0.14.1 Integers

Basic Definitions and Theorems

0.15 Modular Arithmetic

0.15.1 Modular Arithmetic

Basic Definitions and Theorems

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Algebraic Geometry

0.16 Algebraic Geometry

0.16.1 Preliminary Definitions

Definition 0.16.1 (Ring). A set R with two binary operations $+: R \times R \rightarrow R$ and $\cdot: R \times R \rightarrow R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.

2. **Associativity of multiplication:**

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$

3. **Multiplicative identity:** There exists $1 \in R$ such that

$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$

4. **Distributivity:**

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Definition 0.16.2 (Commutative Ring). A ring R is *commutative* if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Polynomial Rings

Definition 0.16.3 (Polynomial). Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression

$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \dots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Definition 0.16.4 (Degree). Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R . If $a_n \neq 0$, then the *degree* of f is

$$\deg(f) := n.$$

The coefficient a_n is called the *leading coefficient* of f . A polynomial with leading coefficient 1 is called *monic*.

Definition 0.16.5 (Polynomial Ring). Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x , denoted $R[x]$, is the set of all polynomials in x with coefficients in R , equipped with addition and multiplication defined by:

$$\begin{aligned} \left(\sum_i a_i x^i \right) + \left(\sum_i b_i x^i \right) &:= \sum_i (a_i + b_i) x^i, \\ \left(\sum_i a_i x^i \right) \cdot \left(\sum_j b_j x^j \right) &:= \sum_k \left(\sum_{i+j=k} a_i b_j \right) x^k. \end{aligned}$$

Polynomial Rings in Several Variables

Definition 0.16.6 (Polynomial Ring in n Variables). Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R , denoted

$$R[x_1, x_2, \dots, x_n],$$

is defined inductively by

$$R[x_1, \dots, x_n] := R[x_1, \dots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_{\alpha} a_{\alpha} x^{\alpha},$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_{\alpha} \in R$, and $x^{\alpha} := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Definition 0.16.7 (Monomial). A *monomial* in $R[x_1, \dots, x_n]$ is a polynomial of the form

$$x^{\alpha} = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Definition 0.16.8 (Total Degree). The *total degree* of the monomial x^{α} is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \dots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Ideals

Definition 0.16.9 (Ideal). Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Definition 0.16.10 (Generated Ideal). Let R be a commutative ring and let $f_1, \dots, f_m \in R$. The *ideal generated by $f_1, \dots, f_m$* is

$$\langle f_1, \dots, f_m \rangle := \left\{ \sum_{i=1}^m r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Definition 0.16.11 (Finitely Generated Ideal). An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \dots, f_m \in R$ such that $I = \langle f_1, \dots, f_m \rangle$.

Definition 0.16.12 (Noetherian Ring). A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Theorem 0.16.13 (Hilbert Basis Theorem). *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \dots, x_n]$ is Noetherian for any field k . Go to proof.

Quotient Rings

Definition 0.16.14 (Quotient Ring). Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Proofs

Capstone

Linear Algebra

0.17 Vector Space Preliminaries

Definition 0.17.1 (List). Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Definition 0.17.2 ($\mathbf{F}^n$). Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \dots, x_n) : x_1, \dots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Definition 0.17.3 (Addition in $\mathbf{F}^n$). Let

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \dots, x_n + y_n).$$

Theorem 0.17.4 (Addition Is Commutative in $\mathbf{F}^n$). For all $x, y \in \mathbf{F}^n$,

$$x + y = y + x.$$

Go to proof.

Definition 0.17.5 (Zero Vector in $\mathbf{F}^n$). The symbol 0 denotes the list of length n all of whose coordinates are 0 :

$$0 = (0, \dots, 0) \in \mathbf{F}^n.$$

Definition 0.17.6 (Additive Inverse in $\mathbf{F}^n$). Let

$$x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \dots, -x_n).$$

Definition 0.17.7 (Scalar Multiplication in $\mathbf{F}^n$). Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \dots, \lambda x_n).$$

Definition 0.17.8 (Vector Space). A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \rightarrow V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

- (i) $u + v \in V$;
- (ii) $u + v = v + u$;
- (iii) $(u + v) + w = u + (v + w)$;
- (iv) there exists $0 \in V$ such that $v + 0 = v$;
- (v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;
- (vi) $av \in V$;
- (vii) $a(u + v) = au + av$;
- (viii) $(a + b)v = av + bv$;
- (ix) $a(bv) = (ab)v$;
- (x) $1v = v$.

Definition 0.17.9 (Vector and Point). An element of a vector space V is called a **vector**. When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Definition 0.17.10 (Real Vector Space). A **real vector space** is a vector space over $\mathbb{R}$.

Definition 0.17.11 (Complex Vector Space). A **complex vector space** is a vector space over $\mathbb{C}$.

Theorem 0.17.12 (Unique Additive Identity). *A vector space has a unique additive identity. Go to proof.*

Theorem 0.17.13 (Unique Additive Inverse). *Every vector in a vector space has a unique additive inverse. Go to proof.*

Theorem 0.17.14 ($0v = 0$). *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Theorem 0.17.15 ($a0 = 0$). *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Theorem 0.17.16 ($(-1)v = -v$). *For every $v \in V$,*

$$(-1)v = -v.$$

Go to proof.

Definition 0.17.17 (Subspace). Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V .

Theorem 0.17.18 (Conditions for a Subspace). *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

- (i) $0 \in U$;
- (ii) for all $u, v \in U$, one has $u + v \in U$;
- (iii) for all $a \in \mathbf{F}$ and all $u \in U$, one has $au \in U$.

Go to proof.

Definition 0.17.19 (Sum of Subspaces). Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V . The **sum** of $U_1, \dots, U_m$ is the set

$$U_1 + \dots + U_m = \{u_1 + \dots + u_m : u_1 \in U_1, \dots, u_m \in U_m\}.$$

Theorem 0.17.20 (Sum of Subspaces Is the Smallest Containing Subspace). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V . Then $U_1 + \dots + U_m$ is a subspace of V containing each of $U_1, \dots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \dots, U_m$, then*

$$U_1 + \dots + U_m \subseteq W.$$

Go to proof.

Definition 0.17.21 (Direct Sum). Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that

$$V = U_1 + \dots + U_m.$$

Then V is called the **direct sum** of $U_1, \dots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \dots + u_m, \quad u_1 \in U_1, \dots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \dots \oplus U_m.$$

Theorem 0.17.22 (Condition for a Direct Sum). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that*

$$V = U_1 + \dots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \dots \oplus U_m$;

(ii) if $u_1 \in U_1, \dots, u_m \in U_m$ and

$$u_1 + \dots + u_m = 0,$$

then

$$u_1 = \dots = u_m = 0.$$

Go to proof.

Theorem 0.17.23 (Direct Sum of Two Subspaces). *Let U and W be subspaces of a vector space V . Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Proofs

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Algebra

Lattice Order

0.18 Breadcrumb

Why Lattice Theory Gets Its Own Home

0.18.1 Ordered Sets

Definition 0.18.1 (Partial Order). A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \implies a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \implies a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Definition 0.18.2 (Strict Partial Order). A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \implies \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \implies a < c$.

Definition 0.18.3 (Total Order). A **partial order** $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \quad a = b, \quad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

0.18.2 Chains and Antichains

Definition 0.18.4 (Chain). Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \vee y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a **total order** on C .

Definition 0.18.5 (Antichain). Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \implies \neg(x \leq y) \wedge \neg(y \leq x).$$

0.18.3 Partial and Total Maps

Definition 0.18.6 (Partial Map). Let A and B be sets. A *partial map* $f : A \rightarrow B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A, \forall y_1, y_2 \in B, (x, y_1) \in f \wedge (x, y_2) \in f \implies y_1 = y_2.$$

Definition 0.18.7 (Total Map). A *partial map* $f : A \rightarrow B$ is a *total map* if it is defined on every element of A :

$$\forall x \in A, \exists y \in B, (x, y) \in f.$$

A total map is written $f : A \rightarrow B$ and is precisely a function in the usual sense.

0.18.4 Maps Between Ordered Sets

Definition 0.18.8 (Order-Preserving Map). Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \rightarrow Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Definition 0.18.9 (Order-Embedding). Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \rightarrow Q$ is an *order-embedding* if

$$\forall x, y \in P, x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Definition 0.18.10 (Order-Isomorphism). An *order-embedding* $\varphi : P \rightarrow Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Lemma 0.18.11 (Every Order-Embedding Is Injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \rightarrow Q$ be an order-embedding. Then φ is injective. Go to proof.*

0.18.5 The Covering Relation

Definition 0.18.12 (Covering Relation). Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x , written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y := x < y \wedge \neg(\exists z \in P, x < z < y).$$

Lemma 0.18.13 (Characterizations of Order-Isomorphisms for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \rightarrow Q$ be a bijection. The following are equivalent:*

1. φ is an order-isomorphism.
2. For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.
3. For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.

Go to proof.

Proposition 0.18.14 (Hasse Diagram Characterization for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements. Go to proof.*

0.18.6 Duality

Definition 0.18.15 (Dual Ordered Set). Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P , denoted P^{op} , is the set P equipped with the reversed order:

$$x \leq_{P^{\text{op}}} y \iff y \leq_P x.$$

Proposition 0.18.16 (Duality Principle). *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op} , obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set. Go to proof.*

0.18.7 Extremal Elements

Definition 0.18.17 (Bottom Element). Let $(P, \leq)$ be a partially ordered set. An element $\perp \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \perp \leq x.$$

If a bottom element exists it is unique; it is denoted $\perp_P$ or simply $\perp$.

Definition 0.18.18 (Top Element). Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Definition 0.18.19 (Maximal Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in Q* if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\text{Max}(Q)$.

Definition 0.18.20 (Minimal Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in Q* if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Definition 0.18.21 (Greatest Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q :

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Definition 0.18.22 (Least Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q :

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Proposition 0.18.23 (Every Nonempty Finite Poset Has a Maximal Element). *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \text{Max}(P)$ with $x \leq y$. Go to proof.*

0.18.8 Lifting and Co-Lifting

Definition 0.18.24 (Lifting). Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp := P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \quad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new **bottom element** $\perp$ adjoined.

Definition 0.18.25 (Co-Lifting). Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \quad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new **top element** $\top$ adjoined.

0.18.9 Consistency

Definition 0.18.26 (Consistency). Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \wedge y \leq z.$$

0.19 Ordered Sets

0.19.1 Down-Sets and Up-Sets

Definition 0.19.1 (Down-Set). Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q, \forall y \in P, \quad y \leq x \implies y \in Q.$$

Definition 0.19.2 (Up-Set). Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q, \forall y \in P, \quad x \leq y \implies y \in Q.$$

0.19.2 Families of Sets

Proposition 0.19.3 (Predicates and Power Sets Are Order-Isomorphic). *The map $\Phi : (\mathcal{P}^*(X), \leq) \rightarrow (\mathcal{P}(X), \subseteq)$ is an *order-isomorphism*. Go to proof.*

Proofs

Capstone